Dataset 2 — Curvigram and sensitivity analysis

File

`02_megalithic_curvigram.csv`

Status

Synthetic teaching dataset.

The dataset was created specifically for the SEAC 2026 workshop. It does not contain measurements from real monuments and must not be cited as empirical archaeological evidence.

Teaching scenario

The file represents **45 hypothetical megalithic monuments** for which field orientations have already been transformed into astronomical declination.

The participants do not need to perform the azimuth/horizon-to-declination transformation in this exercise. The statistical starting point is the already calculated declination plus its reported measurement uncertainty.

Variables

Variable	Meaning
`monument_id`	Synthetic monument identifier.
`declination_deg`	Astronomical declination in degrees. This is the value to be smoothed in the curvigram.
`uncertainty_deg`	Reported ± measurement uncertainty in declination degrees. It is not automatically assumed to be one standard deviation.
`monument_type`	Synthetic monument type used for archaeological context.
`setting`	Simplified synthetic landscape/topographic setting.
`context_note`	Reminder that field context can influence measurement and interpretation.

Why declination?

Declination is used here because the workshop's curvigram lesson is about:

- measurement uncertainty;
- kernel choice;
- bandwidth;
- smoothing;
- sensitivity analysis.

Declination is treated as a linear quantity over the analysed range, so participants do not have to learn 0°/360° circular boundary handling at the same time.

Deliberate statistical structure

The dataset contains:

- one **strong, robust concentration** near +23°;
- one **smaller, less robust feature** near +16°;
- a dispersed background.

Participants are **not told which observations generated which feature**.

The purpose is to discover the structure through the curvigram and then see which features survive reasonable changes in smoothing.

The main concentration is near a declination that may tempt participants to attach an immediate astronomical label. The exercise should explicitly discourage that shortcut:

> a peak is a statistical concentration first; an archaeological/astronomical interpretation requires additional evidence.

Reported uncertainty

For example:

```
declination_deg = 23.4
uncertainty_deg = 0.6
```

means the transformed declination is reported approximately as:

```
+23.4° ± 0.6°
```

The workshop does not assume that ±0.6° is one standard deviation unless the measurement protocol explicitly defines it that way.

Baseline curvigram

The Phase 4 participant script will start with:

```
Kernel:              Epanechnikov
Bandwidth:           2 × uncertainty_deg
Vertical scaling:    area = 1
Calculation step:    0.05°
Range:               -45° to +45°
```

Required sensitivity analysis

Participants will compare:

1. Epanechnikov, multiplier 1;
2. Epanechnikov, multiplier 2;
3. Epanechnikov, multiplier 3;
4. Gaussian, multiplier 2.

The expected teaching pattern is:

- the major feature should remain near +23° throughout;
- the smaller feature should become progressively less distinct under stronger/smoother specifications.

Instructor-only note

The true synthetic generation groups are stored separately in:

`03_SOLUTIONS/02_megalithic_curvigram_GENERATION_KEY.csv`

That file should not be distributed as part of the participant exercise.